

Simple Interest

$$F = P + I = P(1 + i)$$

$$I = P \cdot i, I_s = N \times P \cdot i$$

Compound Interest

$$F = P(1 + i)^N$$

$$I_c = F - P = P(1 + i)^N - P$$

effective rate

derived from
made up of
integer multiples
of i_s

$$i_e = (1 + i_s)^M - 1$$

$$i_s = \frac{r}{m} \quad \begin{array}{l} \text{nominal rate} \\ m \text{ equal} \\ \text{compounding} \\ \text{periods} \end{array}$$

continuous compounding

$$i_e = e^r - 1$$

SLD

$$D_{SL}(n) = \frac{P - S}{N}$$

$$BV_{SL}(n) = P - n \left(\frac{P - S}{N} \right)$$

DBM

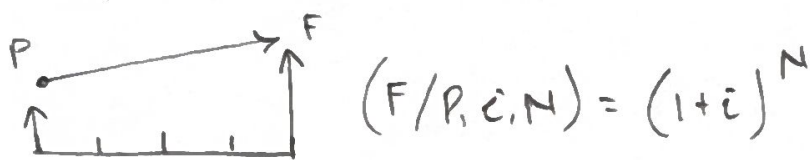
$$BV_{DB}(0) = P$$

$$P_{DB}(n) = BV_{DB}(n-1) \times d$$

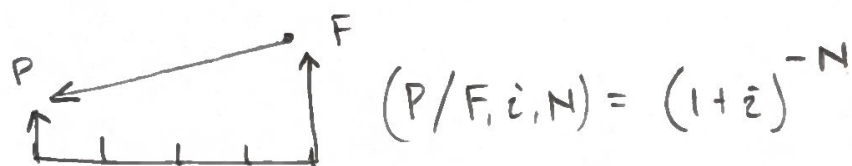
$$BV_{DB}(n) = P(1 - d)^n$$

$$d = 1 - \left(\frac{S}{P} \right)^{1/N}$$

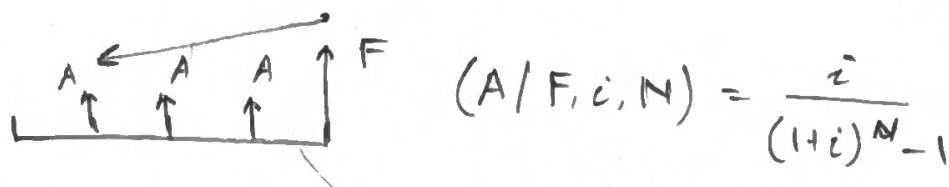
Compound amount factor



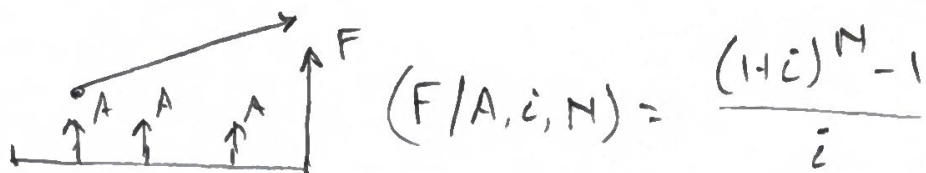
present worth factor



sinking fund factor



uniform series compound amount factor



what is a bond worth today?

present value of face value + present worth of coupons

$$(\sim) (P/F, i, N)$$

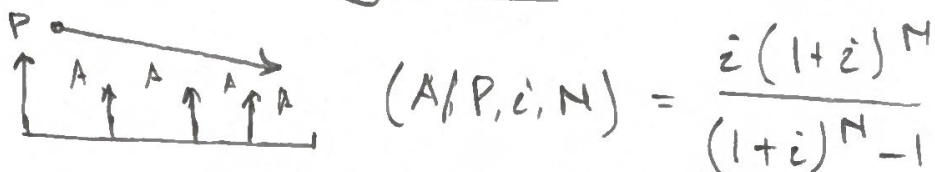
face value

future
value

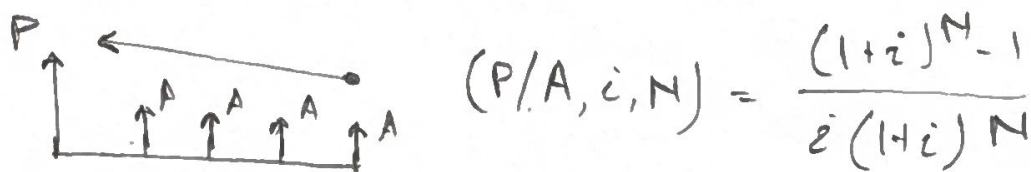
$$+ (\sim) (P/A, i, N)$$

(coupon rate) $\left(\frac{\text{face value}}{\text{comp periods/yr.}} \right)$

capital recovery factor



series present worth factor



capital recovery formula

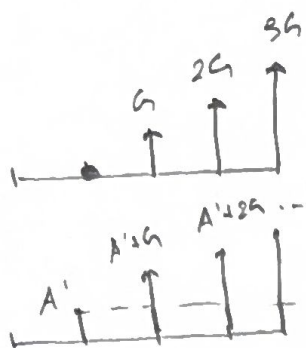
or

capital recovery cost

$$A = (P - S)(A/P, i, N) + iS$$

initial cost P, salvage value S
of t_u N periods @ i

Arithmetic Gradient Series



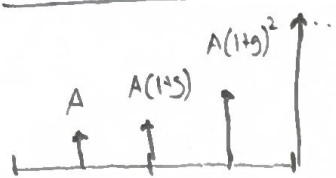
$$(A/G, i, N) = \frac{1}{i} - \frac{N}{(1+i)^N - 1}$$

$$A_{\text{tot}} = A' + G(A/G, i, N)$$

Non-std Annuities/Gradients

capitalized
value formula: $P = A \div i$

Geometric Gradient Series



growth-adjusted
interest rate:

$$i^0 = \frac{1+i}{1+g} - 1 \quad \text{st.} \quad \frac{1}{1+i^0} = \frac{1+g}{1+i}$$

$$(P/A, g, i, N) = \frac{(1+i^0)^N - 1}{i^0 (1+i^0)^N} \left(\frac{1}{1+g} \right) = \frac{(P/A, i^0, N)}{1+g}$$

Cases:

1) $i > g > 0$: $i^0 > 0 \rightarrow$ table/Formula

2) $i < 0$: $i^0 > 0 \rightarrow$ "

3) $g > i > 0$: $i^0 < 0 \rightarrow$ Formula

4) $g = i > 0$: $i^0 < 0 \rightarrow P = N \left(\frac{A}{1+g} \right)$